Technical Document #698: Data Engineering - Comprehensive Overview

Technical Document #698: Data Engineering - Comprehensive Overview

Stream processing handles continuous data streams in real-time. Apache Kafka and Apache Flink are popular stream processing frameworks.

ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a target data store.

Data lakes store raw data in its native format. They support structured, semi-structured, and unstructured data at any scale.

Schema evolution handles changes in data structure over time. Schema registries manage schema versions and ensure backward compatibility.

Partitioning divides data into smaller, more manageable pieces. It improves query performance by reducing the amount of data scanned.

Data quality ensures data is accurate, complete, and consistent. Data validation, cleansing, and monitoring are essential components.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Key Takeaways:

- Data quality ensures data is accurate, complete, and consistent.
- Data governance establishes policies and procedures for managing data assets.
- Data warehouses store structured data optimized for analytical queries.